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Magnetized white dwarfs with mixed poloidal-toroidal fields

Draft – structure and core physics only

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ABSTRACT

Context. TBD.
Aims. TBD.
Methods. TBD.
Results. TBD.

Key words. white dwarfs – stars: magnetic field – magnetohydrodynamics (MHD) – methods: numerical

1. Introduction

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2. Equation of state

The background stellar structure throughout this work is that of a fully degenerate, zero-temperature electron gas (ztwd; Chandrasekhar 1935). Pressure and density are both parametrized by the normalized Fermi momentum $x \equiv p_F/(m_e c)$,

$$P(x) = A\left[x(2x^2-3)\sqrt{x^2+1} + 3\sinh^{-1}x\right], \tag{1}$$

$$\rho(x) = B\,x^3, \tag{2}$$

where A is a fixed combination of fundamental constants (m_e, c, $\hbar$) and B depends on the composition only through the mean molecular weight per electron $\mu_e = 1/Y_e$, with Y_e the number of electrons per baryon. This is a purely mechanical equation of state: at fixed composition, pressure depends on density alone, with no temperature dependence. Consequently, once the configuration is restricted to be non-rotating and field-free, the entire stellar structure (mass, radius, density profile) is determined by exactly two physical parameters: the central density ρ_c and μ_e.

The specific enthalpy has a closed form,

$$H(x) = \frac{8A}{B}\left[\sqrt{1+x^2} - 1\right], \tag{3}$$

which is what makes the self-consistent field (SCF) iteration of Sect. 3 tractable: at each iteration the density is recovered directly from the local enthalpy via the inverse relation $x(H) = \sqrt{(1+HB/8A)^2-1}$, with $H \le 0$ mapped to $\rho = 0$ (the stellar surface).

Near the surface ($x \ll 1$) the equation of state reduces to $\rho \propto H^{3/2}$, an effective polytropic index $n = 3/2$; in the relativistic core ($x \gg 1$) it approaches $\rho \propto H^3$, $n = 3$ – the marginal polytropic index at which the equilibrium mass becomes independent of central density, the origin of the Chandrasekhar mass limit.

At sufficiently high density, inverse beta decay (electron capture) changes the composition locally and the constant-μ_e cold equation of state above ceases to describe a real white dwarf. We adopt the tabulated threshold densities of Boshkayev et al. (2013) for this validity limit; all central densities discussed below are kept below this threshold for the assumed composition.

3. Field equations

Axisymmetric magnetohydrostatic equilibria with a poloidal field are governed by the Grad-Shafranov equation (Grad & Rubin 1958; Shafranov 1966) for the poloidal flux function $u(\varpi, z) = \varpi A_\varphi$, where ϖ is the cylindrical radius and A_φ the azimuthal component of the vector potential. The poloidal field components follow from u as

$$B_\varpi = -\frac{1}{\varpi}\frac{\partial u}{\partial z}, \qquad B_z = \frac{1}{\varpi}\frac{\partial u}{\partial \varpi}, \tag{4}$$

and the equilibrium (force balance combined with the Poisson equation for self-gravity) reduces to a single elliptic equation for u sourced by the current distribution and, self-consistently, by u itself through the poloidal current function $M(u)$.

Following the SCF approach of Hachisu (1986), extended to magnetized configurations, the poloidal current function is taken as linear in the flux, $M(u) = k_0\,u$, so that the Bernoulli integral governing hydrostatic balance reads

$$H + \Phi - M(u) = C, \tag{5}$$

with Φ the gravitational potential and C a constant fixed by the boundary condition at the stellar surface. Equation 5 and the Grad-Shafranov equation are solved iteratively together with the Poisson equation for Φ, alternating between an update of the density field (from the equation of state, Sect. 2) and an update of the flux function u, until convergence.

A toroidal field confined within the last closed poloidal field line ($u > u_c$) is added through a toroidal current function $\beta(u)$, parametrized as $\beta(u) = \zeta(u - u_c)^{m+1}$ with $m \geq 1$, giving $B_\varphi = \beta(u)/\varpi$ inside the closed-field region and zero outside. A purely toroidal, non-mixed branch is also considered separately, with $B_\varphi = K\rho^m \varpi^{2m-1}$ derived directly from the Lorentz force (Maxwell stress tensor) rather than from a flux function; this branch requires no Grad-Shafranov solve, at the cost of being unable to represent a poloidal component. A genuinely self-consistent *mixed* poloidal-toroidal equilibrium – both components simultaneously present and mutually consistent within the same SCF solve – is outside the scope of the axisymmetric formulation used here; Sect. 5 describes how mixed configurations are instead obtained dynamically.

Purely toroidal and purely poloidal equilibria are each subject to a non-axisymmetric instability of azimuthal wavenumber $m = 1$ (Tayler 1973; Markey & Tayler 1973), which the axisymmetric formulation of this section cannot, by construction, test for.

4. Castro

Dynamical (as opposed to SCF-equilibrium) evolution is carried out with Castro (Almgren et al. 2010), a massively parallel, block-structured adaptive-mesh-refinement (AMR) code for compressible astrophysical flows, built on the AMReX framework (Zhang et al. 2019). The hydrodynamics solver uses an unsplit corner-transport-upwind scheme (Colella 1990) with piecewise-parabolic reconstruction (Colella & Woodward 1984); self-gravity is included for isolated (non-periodic) configurations, and a constrained-transport magnetohydrodynamics (MHD) solver extends the same integrator to evolve a magnetic field while preserving $\nabla\cdot\mathbf{B} = 0$ to machine precision by construction. Castro accepts an arbitrary equation of state and reaction network through the StarKiller Microphysics interface; for this work the network is reduced to a single inert species reproducing the assumed μ_e, and the equation of state is the same zero-temperature degenerate electron gas of Sect. 2, so that a star built in hydrostatic equilibrium by the SCF solver is evolved under the identical thermodynamic relation it was constructed with.

Self-gravity for an isolated spherical mass distribution is obtained from Castro's monopole gravity solver, which integrates the enclosed mass along radial shells. Initial conditions are supplied to Castro as a one-dimensional radial density (and composition) profile, sampled onto the three-dimensional Cartesian AMR grid; the sampling itself (cell-volume averaging versus point sampling, and the alignment of the grid relative to the stellar center) has a measurable effect on how closely the discretized initial condition reproduces the target central density, which we account for when initializing a background star for dynamical evolution.

5. The Braithwaite method

As noted in Sect. 3, the axisymmetric SCF formulation cannot produce a genuinely self-consistent mixed poloidal-toroidal equilibrium, nor can it test the stability of any equilibrium against non-axisymmetric perturbations – a serious limitation given that both purely poloidal and purely toroidal fields are themselves non-axisymmetrically unstable (Tayler 1973; Markey & Tayler 1973). The resolution adopted here follows Braithwaite & Spruit (2004) and Braithwaite & Nordlund (2006): rather than constructing a mixed field and separately testing its stability, a *random*, multi-scale magnetic field is seeded within an otherwise field-free, non-rotating background star (Sect. 2) and allowed to relax dynamically under full three-dimensional, ideal MHD in Castro (Sect. 4). Because the evolution is a genuine dynamical calculation and not an imposed equilibrium, any configuration that survives the relaxation is, by construction, stable to the perturbations sampled by the initial random field – including non-axisymmetric ones the Grad-Shafranov formulation cannot represent at all.

The random seed field is constructed as a vector potential $\mathbf{A}$ – a superposition of Fourier-like modes spanning a band of length scales, with random amplitude, phase, and direction for each mode – multiplied by a smooth radial envelope that confines the field within the stellar volume before the magnetic field itself is formed as the discrete curl, $\mathbf{B} = \nabla \times \mathbf{A}$. Windowing the potential rather than the field is required for $\nabla\cdot\mathbf{B} = 0$ to be preserved exactly: multiplying an already-divergence-free $\mathbf{B}$ by a confinement window after the curl has been taken would reintroduce a spurious divergence, whereas curling an already-windowed potential does not. The total seed field energy is calibrated, prior to relaxation, to a chosen fraction of the star's gravitational binding energy, $E_{\rm mag}/|W|$.

Once released, the field is left free to evolve under the full MHD equations; no particular final poloidal-to-toroidal ratio is imposed. The relaxed configuration's toroidal fraction, $E_{\rm tor}/E_{\rm mag}$, is measured directly from the evolved field, and its dependence on the initial random seed (reproducibility across independent realizations of the random field, at fixed $E_{\rm mag}/|W|$) is examined empirically rather than assumed.

6. Results

This section reports the results obtained so far with the procedure of Sect. 5, together with the numerical checks that establish the interval in which they are meaningful. All relaxations share the same non-rotating, initially field-free background star ($\rho_c \simeq 10^9\,{\rm g\,cm^{-3}}$, $\mu_e = 2$), built by the SCF solver of Sect. 3 to a virial error of 1.1×10^{-4} and mapped onto a 64^3 Cartesian grid, with one seed reproduced at 128^3 as a resolution cross-check.

6.1. The measurement window

The MHD path of the code used here is not well-balanced: a star initialized in exact hydrostatic equilibrium does not remain exactly so on the discrete grid, and its central density drifts slowly away from the value it was constructed with. Waiting for the background star to settle is therefore not an available strategy, and the field diagnostics are instead measured inside a *validity window* in time, bounded below by the relaxation of the seed field itself and above by the onset of structural drift of the star.

The lower bound is set at $t/t_{\rm dyn} = 0.4$, by which time the measured toroidal fraction has left its initial transient and entered a quasi-steady band. The upper bound is the first sample at which the central density leaves a $\pm 2\%$ band around its own $t = 0$ value. For the background star used here this gives a window $t/t_{\rm dyn} \in [0.4, 1.13]$, and all measurements reported below fall inside it ($t/t_{\rm dyn} = 0.79$–1.13).

Two independent checks support this choice. First, the same construction at 128^3 never leaves the tighter $\pm 1\%$ band over the entire baseline tested, $t/t_{\rm dyn} = 0.4$–1.45: the window is conservative rather than marginal, and becomes more so with resolution. Second, plotting the toroidal fraction against the central density rather than against time separates the two regimes cleanly

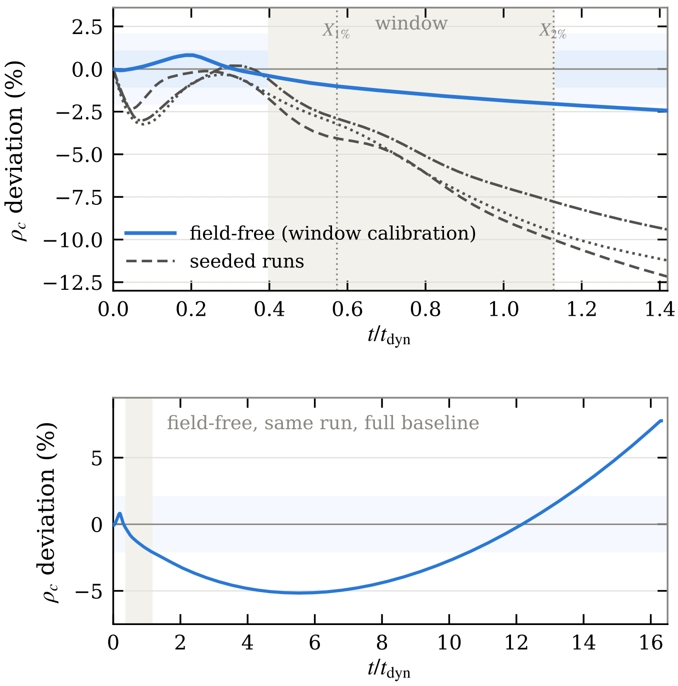

Fig. 1. Central density relative to its initial value. Upper panel: the field-free calibration run (thick) and the three extended seeded runs (thin), with the ±1% and ±2% bands, the crossings $X_{1\%} = 0.573$ and $X_{2\%} = 1.128$ that define the upper bound, and the resulting window shaded. The seeded runs leave the ±2% band well before the window closes. Lower panel: the same field-free run over its full baseline – the drift never settles.

– inside the window the two quantities are essentially uncorrelated, whereas past the upper bound they become strongly correlated ($r \simeq 0.96$–1.00). Beyond the window the measured ratio is slaved to the drifting stellar structure and no longer reflects the dynamics of the field, which is precisely what the upper bound is constructed to exclude.

The lower panel of Fig. 1 shows why waiting is not an option: followed out to $t/t_{\mathrm{dyn}} = 16$, the field-free star does not converge to any value but wanders, from −5% near $t/t_{\mathrm{dyn}} = 5$ back through zero and up to +8% by the end of the run. There is no late-time plateau to wait for.

The upper panel, however, carries a caveat that we consider the most important limitation of the measurements below, and it is visible in the same figure. The window was calibrated on a *field-free* evolution. The seeded runs it is then applied to drift several times faster: at $t/t_{\mathrm{dyn}} = 1.13$, where the field-free star has moved by −2% by construction, the three extended seeded runs have moved by −8% to −10%. The comparison is not an artifact of the damping schedule – at these times both the field-free and the seeded runs are still inside their fully damped phase, before any ramp-off begins – so the difference is attributable to the presence of the field. What it is *not* possible to say from these runs is how much of that extra motion is the star's physical response to magnetic pressure in its core and how much is faster numerical drift. Either way, the operational consequence is the same and we state it plainly: the ±2% criterion bounds the structural change of the calibration run, not of the runs being measured.

6.2. Divergence control

The constrained-transport scheme is expected to preserve $\nabla\cdot\mathbf{B} = 0$ to machine precision, and this is verified on every measurement rather than assumed. The divergence must be evaluated on the interior of each grid patch: computed as a derived quantity without an extra ghost-cell layer, it reads one cell beyond the valid region and returns meaningless values at patch and domain boundaries, which we found to mimic a catastrophic growth of $|\nabla\cdot\mathbf{B}|$ that the field itself does not exhibit. Masking the two cells nearest any boundary, the maximum relative interior divergence stays in the range 5.3×10^{-10}–8.5×10^{-10} at 64^3 and reaches 2.5×10^{-9} at 128^3. The mild increase with resolution is consistent with a finer grid resolving smaller-scale field structure at a fixed precision floor, not with a growing constraint violation.

Table 1. Toroidal fraction of the relaxed field, measured inside the validity window of Sect. 6.1.

Batch	n	$E_{\mathrm{tor}}/E_{\mathrm{mag}}$
A (64^3)	10	0.317–0.370
B (64^3)	10	0.276–0.402
C (64^3)	10	0.280–0.388
Cross-check (128^3)	1	0.305
All	30	0.276–0.402

6.3. The relaxed field is poloidal-dominated

Thirty relaxations were run from independent realizations of the random seed field described in Sect. 5, all calibrated to the same initial field strength $E_{\mathrm{mag}}/|W| = 0.15$: three batches of ten distinct seeds at 64^3, plus one seed repeated at 128^3. Table 1 summarizes the measured toroidal fractions.

Every one of the thirty measurements is poloidal-dominated, $E_{\mathrm{tor}}/E_{\mathrm{mag}} < 0.5$, with no exception in any batch or at either resolution, and the combined band is narrow, $[0.276, 0.402]$. Batch A was run against a background star built before a refinement of the initial-condition sampling and batches B and C against the corrected one; the two constructions give the same band, so the result does not depend on that detail of the mapping. The resolution cross-check is consistent with the 64^3 ensemble: the same seed gives 0.324 at 64^3 and 0.305 at 128^3, a shift small compared to the seed-to-seed spread.

The relaxed configuration is thus mixed and poloidal-dominated, and no poloidal-to-toroidal ratio was imposed to make it so. Both purely poloidal and purely toroidal fields are unstable to non-axisymmetric perturbations (Tayler 1973; Markey & Tayler 1973), and the surviving configurations are neither – consistent with the picture of Braithwaite & Spruit (2004) and Braithwaite & Nordlund (2006), in which a stable stellar field must contain both components. We emphasize that this is a statement about the configuration reached within the validity window and sustained there, not about an asymptotic equilibrium: with the star itself drifting on a longer timescale (Sect. 6.1), the present calculation cannot follow the field to arbitrarily late times.

The spread within the band is a property of the individual seed, not sampling noise. Re-running seeds over a baseline four times longer than the measurement window preserves their rank ordering: had the spread been different realizations sampled at different phases of a common oscillation, extending the baseline would have scrambled that ordering. For this reason we report the ensemble as a band and as the per-seed values, and deliberately not as a mean with an error bar. What sets a given seed's value remains open. The natural candidate – the net magnetic helicity of the initial random field, a quantity conserved under ideal MHD and the standard explanation for the endpoint of this kind

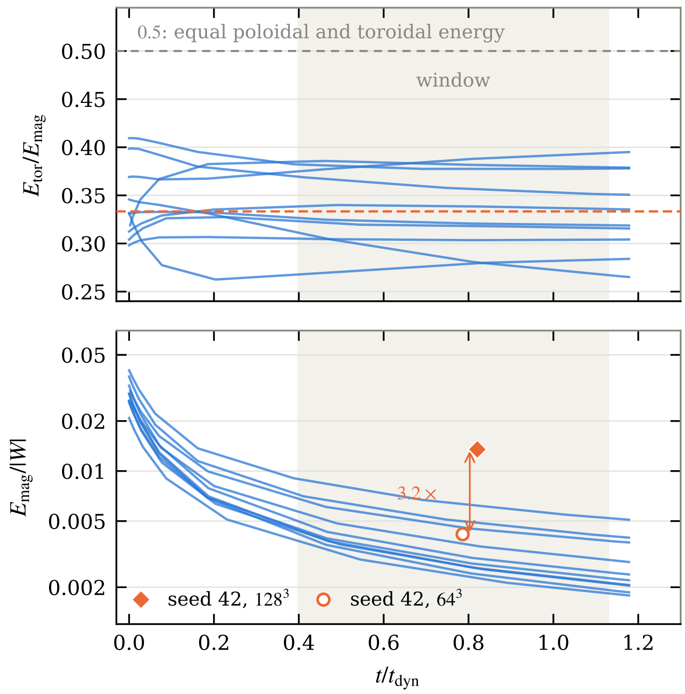

Fig. 2. Evolution of the ten seeds of one batch at 64^3, with the validity window shaded. Upper panel: toroidal fraction, which stays within a narrow band well below 0.5 throughout, with the isotropic value 1/3 marked. Lower panel: magnetic energy, which falls by about an order of magnitude, and the same seed measured at both resolutions – the 128^3 run retains 3.2 times more field energy at the same time, so this decay is resolution-limited.

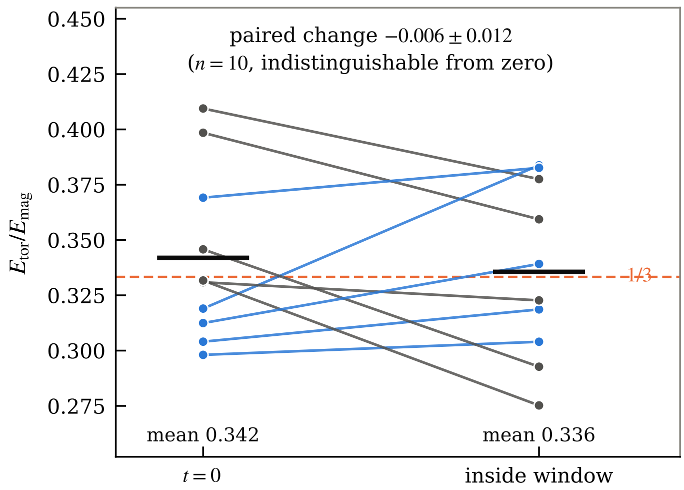

Fig. 3. Toroidal fraction at $t = 0$ and averaged inside the validity window, paired per seed, for the batch of ten. Blue lines increase, gray lines decrease; heavy bars are the ensemble means, and the dashed line is the isotropic value 1/3. Individual seeds move; the ensemble mean does not.

of relaxation – was tested directly and does not survive: a suggestive correlation at $n = 4$ ($r \simeq -0.84$) collapsed to $r \simeq -0.33$, unstable under leave-one-out resampling, once the full batch of ten was included.

6.4. What the relaxation changes, and what it does not

The measurements above are single values, taken inside the window. The plotfiles of one full batch of ten seeds survived, which lets us follow both the ratio and the field energy through the relaxation rather than sampling them once (Fig. 2), and the two behave completely differently.

The toroidal fraction stays inside a narrow band, 0.26–0.41, for the whole evolution. It is not merely poloidal-dominated at the measurement time: it never approaches 0.5 at any point in any run, including $t = 0$. Individual seeds move within the band, in both directions, and settle to their own level well before the window opens.

The magnetic energy, in contrast, falls by roughly an order of magnitude over the same interval, from $E_{mag}/|W| = 0.021$–0.040 at $t = 0$ to 0.0018–0.0051 by $t/t_{dyn} \simeq 1.2$. That decay is resolution-limited, which we can show with the one seed run at both resolutions: measured at essentially the same time, seed 42 retains $E_{mag}/|W| = 0.0135$ at 128^3 against 0.0042 at 64^3, a factor 3.2, while its toroidal fraction changes by only 6% (0.305 against 0.324). The energy left in the field at a given time is therefore a property of the grid as much as of the physics, and we do not report it as a measurement of anything; the ratio is the quantity that survives the resolution change, which is why it is the only one we quote. A convergence study of the decay rate itself would need more than the single 128^3 run available here.

6.5. The toroidal fraction does not move from its seeded value

There is a further consequence of having the full time series, and it bounds the interpretation of the whole ensemble. A statistically isotropic random field puts, on average, one third of its energy into the azimuthal component by symmetry alone. The seed fields are constructed to be random and multi-scale, and they duly start there: for the batch of ten, the toroidal fraction at $t = 0$ averages 0.342, with a seed-to-seed scatter of 0.039. The in-window values average 0.336, with a scatter of 0.039.

Figure 3 pairs the two per seed. Individual seeds move by up to ± 0.065, in both directions, but the paired change across the batch is -0.006 ± 0.012, indistinguishable from zero at $n = 10$, and both ensemble means sit on the isotropic value. Within the window accessible here, the relaxation does not drive the toroidal fraction anywhere: it leaves it, on average, where the random seed put it.

This does not weaken the statement that the surviving configuration is mixed and poloidal-dominated – that is measured, at every time, in every run. It does mean these calculations do not demonstrate the stronger claim they might be read as supporting, namely that the relaxation *selects* a preferred poloidal-to-toroidal ratio. Demonstrating selection requires showing that the field is driven to a ratio it did not start at, and that is exactly what the accessible window is too short to show: with the energy still falling by an order of magnitude per dynamical time and only two to three plotfiles per run falling inside the window, the ensemble mean has neither the time nor the sampling to move. Extending the baseline requires closing the well-balancing gap of Sect. 6.1 first, not running the present setup for longer.

6.6. The spatial structure of a relaxed field

The diagnostics above are volume integrals. Figures 4 and 5 show what the field they integrate over actually looks like, for seed 20 at $t/t_{dyn} = 0.83$ – a plotfile inside the validity window, from the realization whose toroidal fraction (0.338) is the closest in the batch to the ensemble mean, so the figure is representative rather than selected.

In the meridional cut (Fig. 4) the field fills the star rather than concentrating in a core or a shell, reaching 1.9×10^{13} G at its peak, and it is structured on scales well below the stellar radius – the seed was multi-scale and the relaxation has not organized it into a single large-scale structure on the timescale accessible here. The total and poloidal panels are nearly indistinguishable, which is what a toroidal fraction of $1/3$ looks like pointwise. The toroidal panel vanishes along the vertical axis: that is geometric, since $B_\varphi \to 0$ as $\varpi \to 0$ by construction, and it is also a visual reminder of a caveat that applies to every toroidal fraction quoted in this paper – the background star is non-rotating and the seed field is random, so the z axis about which the decomposition is taken is a bookkeeping choice, not a dynamically preferred direction. For a statistically isotropic field any axis gives one third, which is the same fact reported from the other side in Sect. 6.5.

Figure 5 follows the field lines themselves, integrated through each of the three fields from a common set of starting points, so that the panels differ only by which field is being followed. The connectivity is what the slices cannot show. Lines of the total field are tangled, reversing direction on scales well short of the stellar radius and filling the volume without organizing into anything larger – the signature of a multi-scale seed that the accessible baseline has not had time to rearrange. Lines of the poloidal part are tangled in the same way, in meridional loops. Lines of the toroidal part, by contrast, are closed circles about the z axis, and necessarily so: with $\mathbf{B}_\mathrm{t} = B_\varphi \,\hat{\mathbf{e}}_\varphi$ by construction, its field lines can only be circles. That panel is therefore not evidence of an organized toroidal structure – what it carries is where the toroidal component is strong, and in which sense it circulates, not a topology the relaxation produced.

Two numbers put the visual impression on a scale. Taking the 90th percentile of $|\mathbf{B}|$ inside the star, 5.96×10^{12} G, as a common reference level, the total field exceeds it in 10.0% of the stellar volume by construction, while the poloidal part exceeds it in 5.9% and the toroidal part in 2.1% – a volume ratio of 2.8 between the two components at the same field strength.

And one caveat that the integrated numbers hide. Poloidal dominance is a statement about volume-integrated energy, not a pointwise one: the median $|B_\mathrm{p}|/|B_\mathrm{t}|$ inside the star is 1.65, but $|B_\mathrm{t}|$ exceeds $|B_\mathrm{p}|$ in 32% of the stellar volume. The field is locally toroidal-dominated over a third of the star while remaining poloidal-dominated overall.

Acknowledgements. TBD.

References

Almgren, A. S., Beckner, V. E., Bell, J. B., et al. 2010, ApJ, 715, 1221
Boshkayev, K., Rueda, J. A., Ruffini, R., & Siutsou, I. 2013, ApJ, 762, 117
Braithwaite, J. & Nordlund, Å. 2006, A&A, 450, 1077
Braithwaite, J. & Spruit, H. C. 2004, Nature, 431, 819
Chandrasekhar, S. 1935, MNRAS, 95, 207
Colella, P. 1990, Journal of Computational Physics, 87, 171
Colella, P. & Woodward, P. R. 1984, Journal of Computational Physics, 54, 174
Grad, H. & Rubin, H. 1958, Proceedings of the 2nd UN Conf. on the Peaceful Uses of Atomic Energy, 31, 190
Hachisu, I. 1986, ApJS, 61, 479
Markey, P. & Tayler, R. J. 1973, MNRAS, 163, 77
Shafranov, V. D. 1966, Reviews of Plasma Physics, 2, 103
Tayler, R. J. 1973, MNRAS, 161, 365
Zhang, W., Almgren, A., Beckner, V., et al. 2019, Journal of Open Source Software, 4, 1370

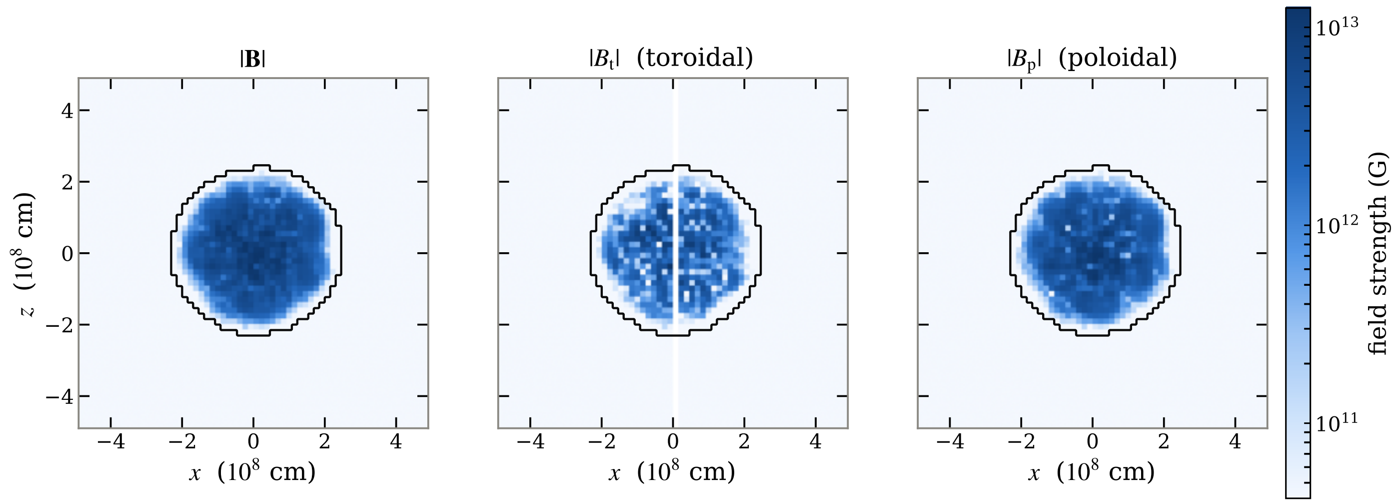

Fig. 4. Meridional cut ($y = 0$) through the relaxed field of seed 20 at $t/t_{\rm dyn} = 0.83$: total field strength, and its toroidal and poloidal parts, on a common logarithmic color scale. The contour is the stellar surface (density at 10^{-3} of peak). The toroidal component vanishes on the axis, where $\varpi \to 0$.

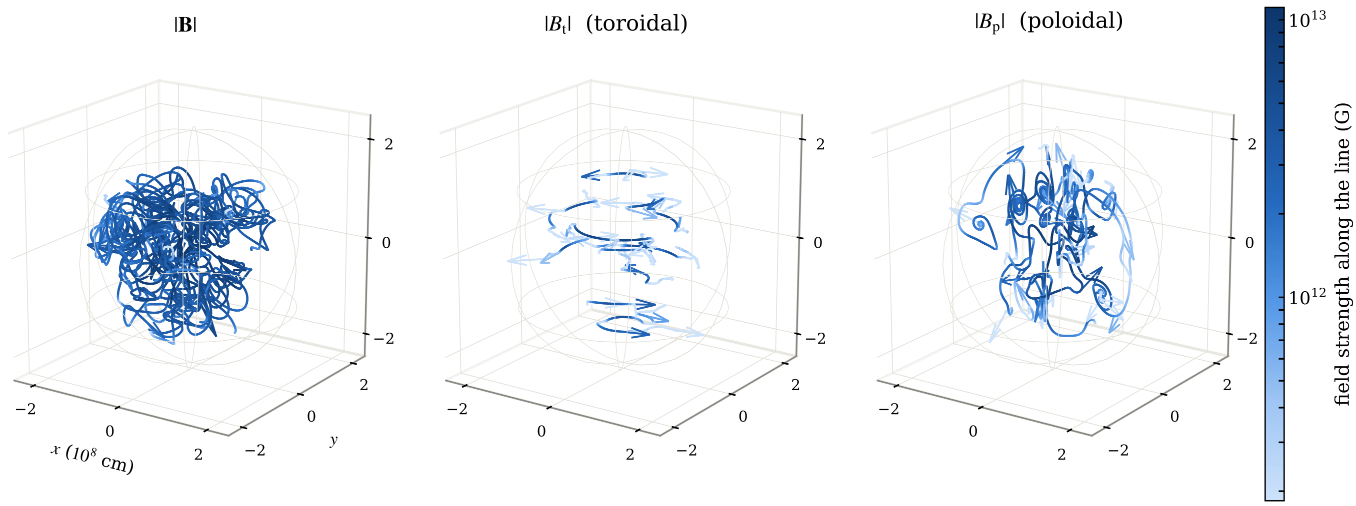

Fig. 5. Field lines of the same three fields as Fig. 4, traced from a common set of starting points in the strong-field region, colored by the local field strength on a shared scale, with arrowheads giving the direction. The faint wireframe is the stellar surface. Lines of the total and poloidal fields are tangled on scales short of the stellar radius; lines of the toroidal component are closed circles about the z axis by construction, so that panel shows where the toroidal field is strong rather than a structure the relaxation selected.